

Basics of Shapes

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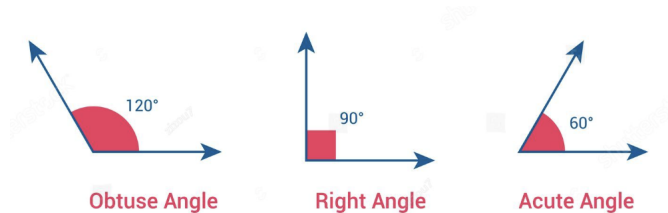
Introduction

I'll talk about some of the formulae you must know about basic geometry. You should be familiar with properties of shapes like triangles, quadrilaterals, and circles.

But before that, let's look at something called angles.

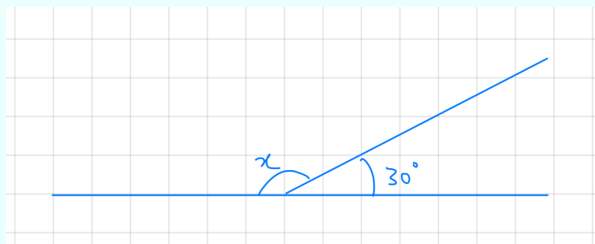
1 Angles

Angles are measured in **degrees**. Depending on the size of the angle, we call them acute, right, and obtuse.



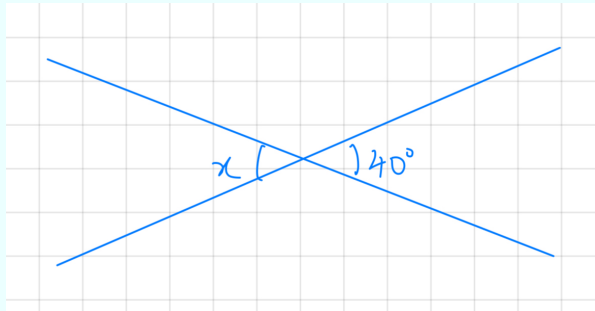
1.1 Angle Rules

Theorem 1.1 Angles on a straight line add to 180°

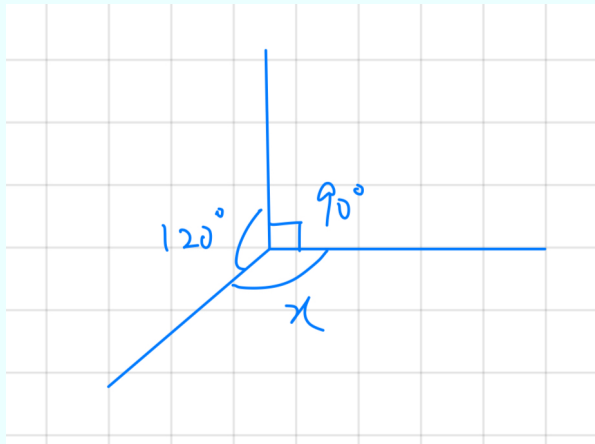


Problem 1.1 What's the size of angle x ?

Theorem 1.2 Vertically opposite angles are equal.



Theorem 1.3 Angles at a point add to 360° .



2 Triangles

2.1 Property 1: Area

The area of the triangle is equal to $\frac{1}{2} \times \text{base} \times \text{height}$.

Problem 2.1 Do you know why?

2.2 Property 2: Angles

The angles inside a triangle will add to 180°

Problem 2.2 I have a triangle. One angle is 3 times as large as the smallest angle, and another angle is 5 times as large as the smallest angle. What are the sizes of the angles in my triangle?

3 Quadrilaterals

4 Circles